← Back **Module Five Quiz**
Graded Assignment • 30 min

Due Mar 20, 11:59 PM CET

1. Which of the following is a true statement concerning applications of phylogenetic analysis? — 1 point

- Phylogenetic analysis of modern corn species has shown that there are currently no living species of the presumed ancestral plant teosintes.
- Sequencing of viruses provides a valuable tool in the identification of emerging disease agents.
- Since phylogenetic analysis does not apply to snake venoms and we know little about the venoms of many species of snakes, anti-venoms must be produced for all species of venomous snakes.
- Phylogenetic analysis is not yet able to provide information that helps scientists understand the timing and direction of the spread of human disease during epidemics.

2. Scientists conducting research at Rusinga Island in Kenya: — 1 point

- focused their efforts on uncovering clues to the environments in which early apes of the genus *Proconsul* were found in order to understand if it was a "generalist" capable of living in different environments
- have identified the region as the place of origin of the first Neanderthal humans
- found that the *Proconsul* seemed unable to cope with environmental change and inferred that *Proconsul* was a "specialist" with a very restricted range and habitat
- found ancestral ape fossils of the genus *Proconsul* that have now been dated to 100 million years ago

3. *Homo sapiens* or "anatomically modern humans": — 1 point

- has, as its most closely *living* relative, the old world monkeys of Africa
- belongs to a clade of related species that evolved during the Pliocene and the Pleistocene in Africa and Eurasia
- is phylogenetically found at the terminus of the monkey branch of the tree of life
- have been shown phylogenetically to have evolved from apes like gorillas and chimps

4. Which of the following is NOT true of the evolution of the primate branch of the Tree of Life? — 1 point

- The great apes include the gibbons, orangutans, gorillas, chimpanzees, and humans.
- Living primates are a well-defined and familiar group of mammals that includes lemurs, tarsiers, Old and New World monkeys, orangutans, gorillas, and chimpanzees but *does not* include modern humans, which are classified separately

  as hominids.
- Lemurs underwent an amazing radiation on the island of Madagascar, while lorises inhabited forests in Southeast Asia, India, and Africa.
- There are approximately 50 species of New World monkeys distributed in the forests of South and Middle America.

5. As scientists study the early origins of our "family tree" they have found: — 1 point

- The fossil remains of the earliest hominids (dating to 6 myr) indicate that man had not yet shifted to bipedal locomotion at that time.
- Modern humans are placed with the Hominids, which include our closest relatives, the chimps and bonobos.
- Early hominid fossils such as *Sahelanthropus tchadensis*, and *Orrorin tugenensis* (dating to 6 myr) show anatomical features that differ from other ancestral apes and show the beginning of divergence of hominids.
- That the diverse group of hominids, the australopithecines are easily defined and can be confidently placed in a monophyletic clad.

6. The fossil evidence for the evolution of hominids: — 1 point

- is unclear due to the fact that there have been no fossil of early hominids found in North America and South America
- supports *H. sapiens* as the single lineage of hominids on the primate phylogenetic tree
- is a mystery due to the lack of hominid fossils younger than 7 million years
- supports the origin of several African species that, it is postulated, proliferated because of geographic isolation produced by repeated climate change events

7. All of the following statements are true regarding modern human evolution EXCEPT: — 1 point

- *Homo sapiens* first appeared in Africa about 200,000 years ago and all people living in the world today can be traced back to an ancestral population that lived in eastern and southern Africa.
- Beginning about 1 million years ago, evidence indicates a clade of four morphologically advanced species in the genus *Homo* evolved from African ancestral species. One of these species was *H. sapiens*.
- Evidence for genetic exchange between *H.sapiens* and the *H. neanderthalensis* sheds doubt on species distinctness between modern humans and the Neanderthals.
- The fossil record indicates *H. heidelbergensis*, a species widespread in eastern and southern Africa by 600,000 years ago, made its way to Eurasia by 500,000 years ago and may be ancestral to both *H. neanderthalensis* and *H. sapiens*.

8. To demonstrate relationships among modern people of the world: — 1 point

- Scientists use the present day geographical distribution and skin types of people of various races to document various ancestral lineages.
- Scientists have used genetic, archaeological, and linguistic evidence to document the migration of *H. sapiens* into Alaska and have found evidence that they crossed the Pacific ocean in Viking-like long boats.
- Scientists use maternal mitochondrial DNA.
- Scientists have tabulated forensic data from medical records to document the migration routes of people into the Americas.

9. Evidence indicates that the development of modern language most likely occurred first in: — 1 point

- a species of *Proconsul* apes
- *H. erectus*
- *H. sapiens*
- *H. rudolfensis*

10. Evidence indicates that the last part of the world to be populated by modern humans was: — 1 point

- the islands of the Pacific Ocean
- the western lands of Europe including Spain and Britain
- the North American region of Alaska
- South America